

# Research Assignment

1. A database is an organised collection of data that can be stored, accessed, updated and managed efficiently. Databases are important because they allow organisations to store large amounts of information securely and retrieve it quickly.
2. A data warehouse is a central system that collects data from different sources for reporting and analysis. Data warehouse is designed mainly for analysing historical data and supporting business decisions.
3. A data lake is a storage system that keeps large amounts of raw data in its original form. It can store structured, semi-structured and unstructured data.
4. Data mart is a smaller part of a data warehouse that focuses on a specific department or business area. e.g. A company may have a sales data mart.
5. **Structured** - Data organised into fixed rows and columns.  
e.g. Excel spreadsheet or SQL table.

**Semi-Structured** - Data that has some organisation but does not follow a fixed table structure.  
e.g. JSON or XML

**Unstructured** - Data without a fixed data structure.  
e.g. images, videos or emails.



6. Data modeling is the process of designing how data will be organised, stored and related to other data.

It is important because it helps create accurate, efficient and understandable data systems.

eg **Relational model** - stores data in tables made up of rows and columns.  
**Document model** - stores data as documents, often in JSON-like structures.

7. Data mining is the process of examining large amounts of data to find useful patterns, relationships and trends.

Business use data mining to understand customers, improve operations and support decision making.

eg - A supermarket can analyse purchasing patterns to identify products frequently bought together.

- A bank can analyse transaction patterns to identify possible fraudulent activities.

8. A Database Management System (DBMS) is a software used to create, store, organise, retrieve and manage data in databases.

| DBMS            | Description                                                  | Suitable use                                    |
|-----------------|--------------------------------------------------------------|-------------------------------------------------|
| MySQL           | Open-source relational database system.                      | Websites & Online applications.                 |
| PostgreSQL      | Open-source relational database with advanced features.      | Complex business applications.                  |
| Oracle Database | Enterprise relational database system.                       | Large organisations & financial systems.        |
| MongoDB         | Document-based database using flexible, JSON-like documents. | Applications with changing or non-uniform data. |



9. ETL means Extract, Transform, Load.

**Extract** Data is collected from different sources.

**Transform** The data is cleaned, corrected and converted into a suitable format.

**Load** The transformed data is loaded into a database, data warehouse or other storage system.

ETL is important because it prepares data so that it can be analysed accurately.

10. **CSV** - Simple text file containing rows and columns. To use on simple tabular data and spreadsheets.

**JSON** - Stores data as key-value pairs and nested objects. Use on APIs and web applications.

**Parquet** - Column-based format designed for efficient analytics. Use on large datasets and data analytics.

**Avro** - A row-based format that stores data together with its schema. Data pipelines and data exchange.

11. Artificial Intelligence (AI) is technology that enables computers to perform tasks that normally require human intelligence.

**Narrow AI**: Designed to perform a specific task, such as a voice assistant. e.g. Google Maps route recommendations.

**General AI**: Refers to a theoretical AI system that could perform a wide range of intellectual tasks like a human. e.g. Hypothetical machine capable of learning and performing many different human intellectual tasks.



12. Machine Learning (ML) is a type of AI where computers learn patterns from data and use them to make predictions or decisions.

Traditional programming uses rules + data  $\rightarrow$  output

Machine learning uses data + examples  $\rightarrow$  learned model.

Supervised Learning: Learns from labelled data

Unsupervised learning: Finds patterns in data without labelled answers.

Reinforcement Learning: Learns by receiving rewards or penalties for actions.

13. Deep Learning is a branch of machine learning that uses neural networks with many layers to learn complex patterns.

The relationship is: Artificial Intelligence  $\rightarrow$  Machine learning  $\rightarrow$  Deep learning.

A neural network is made of connected processing units called neurons. It receives data, processes it through layers and produces an output.

14. Natural Language Processing (NLP) is a branch of AI that allows computers to understand, process and generate human language.  
e.g. Siri, Google assistant, Chat GPT

15. Large Language Model (LLM) is a AI model trained on very large amounts of texts so that it can understand and generate human-like language.

Standard machine learning model may be designed for one specific prediction task, while an LLM can perform many language-related tasks.

GPT - used for conversation, writing, reasoning & coding

Claude - used for writing, analysis, coding & other language tasks.



16. Generative AI creates new content such as text, images, audio, video or code. At a high level, it learns patterns from training data & uses those patterns to generate new content based on a user's instructions.

eg. ChatGPT - generates and analyses text.

DALL-E - generates images from descriptions.

GitHub Copilot - assist with writing computer code.

17. Training dataset is a collection of examples used to teach a machine learning model. Good-quality training data is important because the model learns patterns from that data. If the data is incorrect or biased, the model may produce inaccurate or unfair results. eg. When facial recognition systems are trained using an unbalanced dataset, it may perform less accurately for some groups of people.

18. Computer vision is an area of AI that allows computers to process and analyse images and videos. Machines use techniques such as image processing and neural networks to identify objects, patterns and features in visual information. eg. Facial recognition

19. Application Programming Interface allows different software systems to communicate with each other. A simple analogy is a waiter in a restaurant. The waiter takes your order, gives it to the kitchen and brings the results back to you. API performs a similar communication role between software systems. APIs are important they allow applications and data systems to work together.

20. API key is a secret code used to identify or authenticate an application when it accesses an API. It should be stored securely, such as in environment variables or secret-management systems. They should not be posted publicly on Github or in source code. If an API key is exposed, another person may use it & cause unauthorised access or unexpected costs.



21. Representational State Transfer. A REST API allows applications to communicate over the web using HTTP

| Method | Purpose                  |
|--------|--------------------------|
| GET    | Retrieves data           |
| POST   | Sends or creates data    |
| PUT    | Updates or replaces data |
| DELETE | Removes data.            |

22. JSON stands for JavaScript Object Notation. It is a lightweight format used to represent and exchange data. JSON is popular with APIs because it is easy for humans to read and easy for computers to process.

23. Prompt engineering is the process of creating clear specific instructions for an AI model to produce a useful response. It is important because better instructions usually help AI to give more relevant results.

Poor prompt - Tell me about python.

Improved prompt - Explain python to a beginner in simple sentences and give examples of how it is used in data analytics.

24. MAP function applies the same operation to each item in a collection of data

e.g numbers = [1, 2, 3, 4]

result = list (map(lambda x: x \* 2, numbers))

The results:

[2, 4, 6, 8]



25. Cloud computing means using computing resources such as storage, servers and software over the internet instead of managing all the hardware yourself.

| Model | Meaning                                                      | Examples          |
|-------|--------------------------------------------------------------|-------------------|
| IaaS  | Provides infrastructure such as servers and storage.         | AWS - Amazon EC2  |
| PaaS  | Provides a platform for developing and running applications. | Google App Engine |
| SaaS  | Provides complete software applications over the internet.   | Google Workspace. |

26. Version Control System records changes made to files so that users can track and manage different versions. Git is a popular <sup>or</sup> version control system widely used by developers and data professionals.

**Commit:** Saves a set of changes to the project history.

**Branch:** A separate line of development used to work on changes without affecting the main version.

27. Jupyter Notebook is an interactive environment where users can write and run code, view results and add text, equations and visualisations in the same documents. It's commonly associated with Python. It is popular because it makes it easy to analyse data, run code, step-by-step and create charts.

28. Python is a general-purpose programming language that is widely used in data analytics, automation, machine learning & AI. It has a simple syntax and many libraries designed for data work.



Pandas - Data cleaning and analysis

Numpy - Numerical calculations

Matplotlib - Creating charts & visualisations

Scikit-learn - Machine learning

29. Data privacy is the protection of personal information and ensuring that it is collected, stored and used appropriately. GDPR is a European Union data protection law that regulates the processing of personal data. It is applied since 2018 May 25 & is incorporated into the European Economic Area agreement. It can also be applied to organisations outside the EU when they offer goods or services to people in the EU or monitor their behaviour.

30. Data ethics means using data and AI responsibly, fairly & respectfully. eg Using personal information without permission. Violates people's privacy.

Discriminatory AI decisions: An AI system may unfairly disadvantage certain groups.

Selling personal data without proper consent. This can misuse people's private information.

31. Data governance is the system of rules, policies, roles and processes used to manage data properly. Important components include Data quality, Data Security, Data privacy, Data Ownership, Data access controls, Data Standards, Data policies, Compliance.

Good governance helps organisations maintain accurate, secure and trustworthy data.



32. AI bias occurs when an AI system produces unfair or systematically inaccurate results for certain people or groups e.g. AI system used for recruitment that learns from historical hiring data that contains existing discrimination. The system may then favour certain groups over others. Possible consequences include unfair hiring decisions, discrimination and loss of trust.

| 33 Career      | Main Role                                           | Key Skills                            | Common Tools                    |
|----------------|-----------------------------------------------------|---------------------------------------|---------------------------------|
| Data Analyst   | Analyses data and creates reports and dashboards    | SQL, Excel, Statistics Visualisation. | Excel, PowerBI, SQL             |
| Data Scientist | Builds models & finds complex patterns in data      | Python, Statistics, ML                | Python, Pandas, Scikit-learn.   |
| Data Engineer  | Builds & maintains data pipelines & infrastructure  | SQL, Programming, database, cloud     | Python, SQL, Spark, Databricks. |
| AI Engineer    | Builds & deploys AI & machine learning applications | Python, AI, ML, Software development  | Python, TensorFlow, PyTorch     |

34. Business Intelligence is the process of using data, analysis & visualisation to help organisations to make informed decisions.

Microsoft PowerBI - creates interactive reports and dashboards.

Tableau - help users analyse and visualise data.

BI tools turn data into understandable information that business can use for decision-making.

35. Artificial intelligence and Machine learning are most interesting because they show how computers can learn from data perform tasks that normally require human intelligence, I was interested in <sup>how</sup> AI is being used in everyday applications such as chatbots & recommendation systems.



I found the difference between data lakes, data warehouses & databases difficult to understand at first. After the research I learned the differences and can differentiate them now.

37. I know data and AI will become important in my future career. Data analytics can help identify trends, create dashboards and make better business decisions. AI can help automate repetitive tasks, analyse information and improve productivity. In everyday life, AI is already used in search engines, recommendations, navigation and virtual assistants.